

May 29, 2020_m - SKETCH OF ANSWERS

Question 1 (5 points)

A closed economy in which only the goods market exists is described by the following system of equations:

$$C = c_o$$

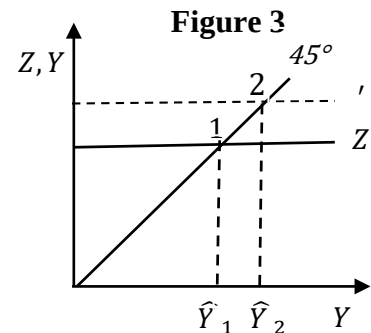
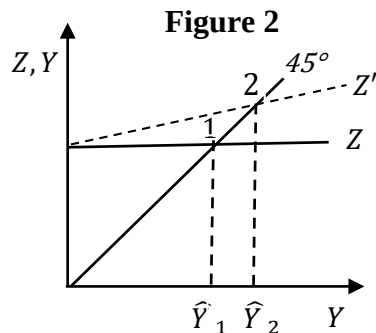
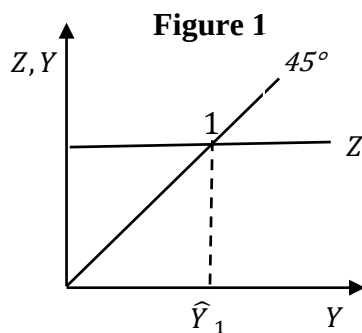
$$I = \bar{I}$$

$$G = \bar{G}$$

$$T = tY$$

where, as always, the positive constant c_o is autonomous consumption, t (a constant between zero and one) is the tax rate, and the other symbols have the usual meaning.

Which of the following three figures provides the correct graphical representation of the impact on equilibrium production, initially equal to $\hat{Y}_1$, of a decrease in the tax rate? Why? Explain.



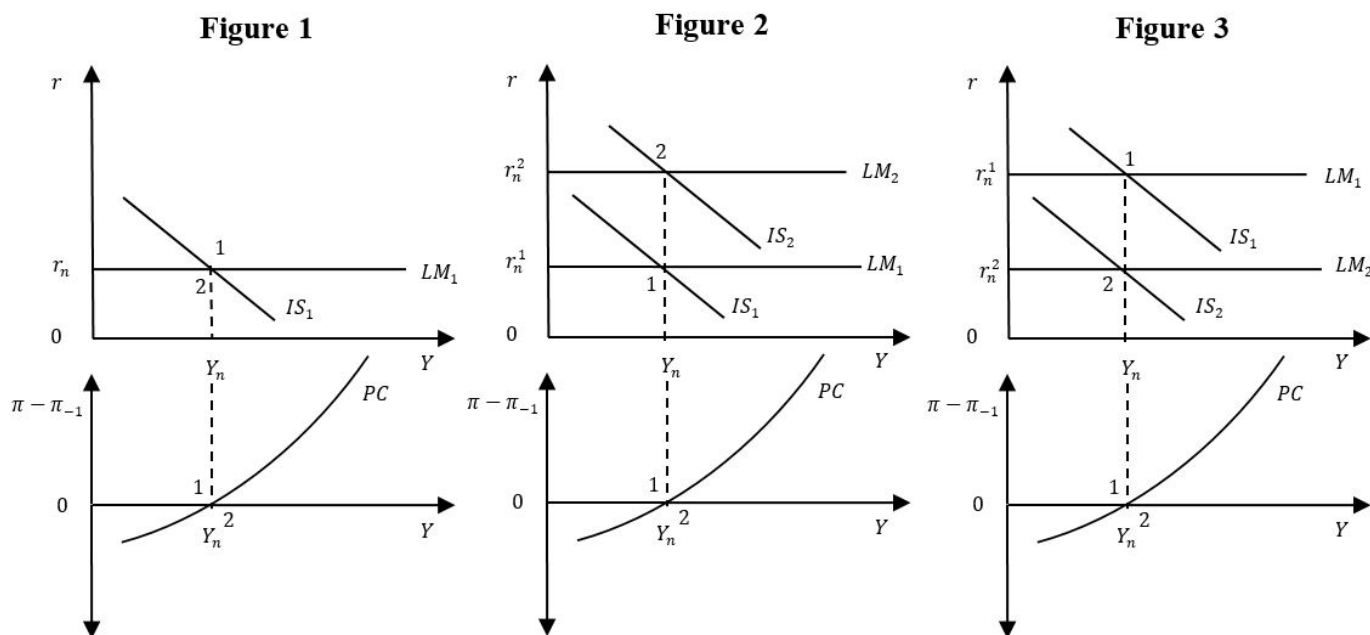
SKETCH OF ANSWER

The correct figure is #1. In fact, in a plane in which Y is measured along the x-axis, the curve representing the aggregate demand for goods is a horizontal line, as all its components are exogenous – in particular, independent of income. For the same reason, a change in t will not cause any change in the graph, and therefore in equilibrium income.

Question 2 (5 points)

Consider a country whose economy is described by an entirely standard IS-LM-PC model, and where the central bank chooses the (real) policy interest rate, r .

Suppose that, when the economy was initially in the medium run equilibrium denoted by '1' in the figures below, the government raises its purchases of goods and services and net taxes by the same amount, so that $\Delta \bar{G} = \Delta \bar{T} > 0$; at the same time, to prevent output from deviating even momentarily from its natural level, the central bank implements an open market operation. Which of the figures below depicts the effects of the policy mix under consideration? By referring to the figure you have chosen, explain whether, to reach its aim, the central bank should purchase or sell bonds, and compare the composition of aggregate demand in the new medium run equilibrium ('2') to that in the initial one ('1').

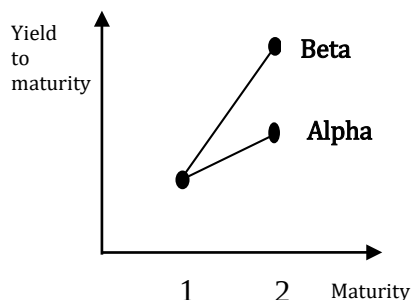


SKETCH OF ANSWER

- *Correct figure: #2.*
- *IS shifts to the right, since the simultaneous increase in G and T raises autonomous spending (by one minus the marginal propensity to consume times the amount by which G – or T – goes up).*
- *To prevent production from rising, the central bank must increase the interest rate, and therefore sell bonds in the open market.*
- *In the equilibrium 2, G is larger than it was at point 1, I smaller (same Y , r higher), C smaller (same Y , T higher).*

Question 3 (5 points)

Alpha and Beta are two closed economies where only one-year and two-year bonds exist. Investors, who do care about risk, ask for a risk premium x to hold the two-year bond. Suppose that the risk premium is smaller in Alpha than in Beta, $x_{Alpha} < x_{Beta}$, and that the yield curves prevailing in the two countries are those represented in the figure below. Using the implications that, for the slope of the yield curve, follow from the approximated relation between yield to maturity of a two-year bond (i_{2t}), the current (i_{1t}) and future expected ($i_{1,t+1}^e$) yields on one-year bonds, and the risk premium (x), explain if from the information on the two economies provided above one can conclude that, in both countries, i_{1t} is necessarily smaller than $i_{1,t+1}^e$. If so, in which of the two countries is the yield on one-year bonds expected to rise the most? If not, why? Explain.

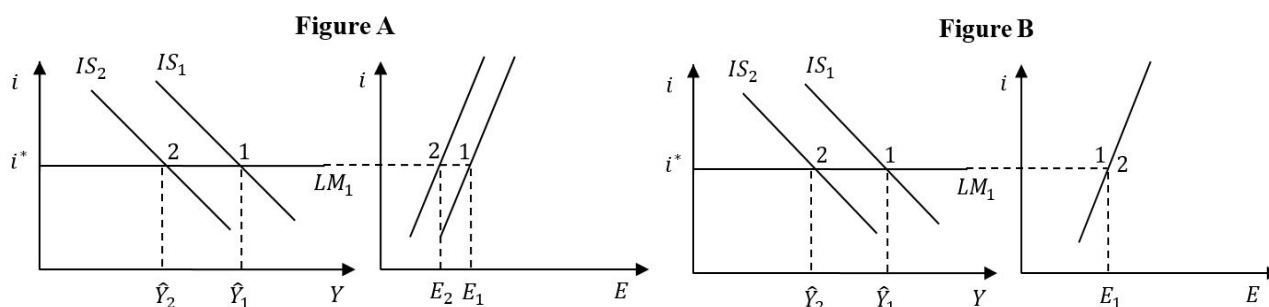


SKETCH OF ANSWER

With risk averse investors, the yield on two-year bonds exceeds the average of the current and future expected yields on 1-year bonds by an amount that is larger the greater is the risk premium, x . It follows that, with risk aversion, a positively sloped yield curve does not necessarily signal expectations of a rise in short term rates (for instance, when $x > 0$ the yield curve will be positively sloped even if investors expect future short rates to be equal to current short rates). Similarly, the fact that Beta's yield curve is steeper than Alpha's could be due only to the larger risk premium prevailing in Beta, rather than to the fact that the yield on one-year bonds is expected to rise more in Beta than in Alpha.

Question 4 (6 points)

Consider Delta, an economy with constant prices under a flexible exchange rate regime described by a standard open economy IS-LM model. In particular, suppose that Delta's central bank, which chooses the domestic interest rate, sets this latter equal to the foreign interest rate, so that $i = i^*$. Suppose that, when Delta was in the initial equilibrium '1' in the figures below, with $i_1 = i^*$, $Y = Y_1$ and $E = E_1$, net taxes and autonomous consumption fall by the same amount, so that $\Delta \bar{T} = \Delta c_0 < 0$. Which of the two pairs of graphs 'open economy IS-LM' - 'interest parity condition' below, the one in Figure A or that in Figure B, describes the effects of these changes on the economy of Delta? In the move from the initial equilibrium ('1') to the new one ('2'), how will the country's equilibrium levels of production, consumption, investment, exchange rate and net exports will have changed in Delta? And what about money supply? Explain.



SKETCH OF ANSWER

- Correct figure: B.
- IS shifts to the left, since – for any given Y – the simultaneous fall in c_0 and net taxes lowers C (to be more precise, C falls by one minus the marginal propensity to consume times the decrease in T – or, which is the same, in c_0) [decrease in C “on impact”, for given Y]
- Since the central bank keeps the interest rate at the initial level, the economy reaches equilibrium 2, where income is lower (due to the fall in demand that has disturbed the initial equilibrium), I lower (because of the decrease in Y), G unchanged, C lower (both due to its decrease on impact, for given Y , mentioned in the second point of this answer, and because of the fall in equilibrium income). Since the interest rate is unchanged, the exchange rate does not vary and – due to the fall in Y , and the ensuing decrease in imports – in the move from 1 to 2 net exports rise.
- As the economy goes from 1 to 2, money demand decreases (i unchanged; Y lower); it follows that money supply will have to fall by the same amount money demand has gone down. The central bank will have to lower M to prevent the decrease in money demand from lowering interest rate below the level it has chosen (i^*).

Question 5 (5 points)

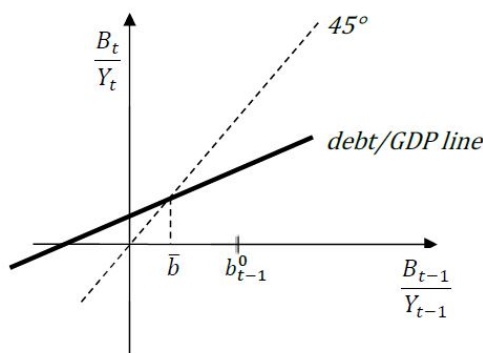
Suppose now that, rather than under a flexible exchange rate, Delta is in a fixed exchange rate regime. In particular, assume that the central bank of Delta wants to keep the exchange rate constant at E_1 , and that market participants expect this exchange rate to prevail also in the future, so that $E = E_1 = E^e$. Starting from the same initial equilibrium '1' described above, net taxes and autonomous consumption fall by the same amount described in the previous question for the case of flexible exchange rates. Now that Delta is in a fixed exchange rate regime, the change in equilibrium production will be larger, smaller or equal to that observed under flexible rates? And what about the change in net exports? Explain.

SKETCH OF ANSWER

Equilibrium income and net exports will vary in the same direction and by the same amount discussed when answering the previous question. The effect of the changes in consideration in an economy under a flexible exchange rate in which the central bank has chosen to set the domestic interest rate equal to the foreign one is identical to that prevailing in an economy under a fixed exchange rate, but otherwise identical. The only difference between the two economies is that, in the second one, the central bank must set $i = i^*$ to keep the exchange rate fixed, while in the first economy a domestic interest rate equal to i^* is just a choice of the central bank.

Question 6 (5 points)

The current time period is t . The dynamics of the debt/GDP ratio for the economy in consideration can be determined on the basis of the figure below, where b_{t-1}^0 is the debt/GDP ratio inherited from the last period. Having explained if, in this country, the economy's growth rate is larger or smaller than the real interest rate, and if the primary budget is in surplus or in deficit, suggest two ways in which policy-makers could stabilize at b_{t-1}^0 the debt/GDP ratio prevailing at time t (that is, two ways in which they could make sure that the debt/GDP ratio will amount to b_{t-1}^0 also at time t). Explain. [Remember that the equation of the debt/GDP line is $b_t = (1 + r - g)b_{t-1} + d$, where $b \equiv B/Y$].



SKETCH OF ANSWER

- Since the debt/GDP line has a positive vertical intercept, $d > 0$ (the government runs a primary deficit); and since the slope of the same line is less than 1, $r < g$.
 - Absent policy interventions, the debt/GDP ratio would fall over time, approaching the steady state level $\bar{b}$ in the figure. To freeze it at b^0 , one has to make b^0 a steady state; in turn, this requires that the debt/GDP line shifts, or rotates, in such a way that it ends up intersecting the dashed line with unit slope going through the origin exactly for $b = b^0$. This can be engineered:
 - for a given slope of the debt/GDP line, by raising its vertical intercept – that is, by increasing the primary deficit, d ;
 - for a given vertical intercept, by increasing the slope of the debt/GDP line – something that could for instance follow from the choice of a higher policy rate r set by the central bank, or from other policy interventions leading to a smaller g ;
- or, finally, through a combination of interventions leading to the changes just described.